

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for enterprise systems, handling over 5,000 requests per minute during peak loads.
- Designed and implemented **Hibernate** entities and complex JPQL queries for data persistence, ensuring data integrity across relational databases.
- Authored unit tests using **JUnit** and Mockito, achieving 85% code coverage for critical backend Java modules.
- Supported the execution of design studies across concept, basic, and detailed design phases for industrial process automation projects.
- Performed process simulations and engineering calculations using **Aspen** and **Mathcad** to support mass and energy balances and equipment sizing.
- Coordinated with **EC&I, Automation, and Mechanical** engineering teams to deliver integrated process solutions, resolving technical queries effectively.
- Containerised backend services using **Docker** and managed deployments on **AWS EC2** instances, improving application portability and scalability.

SKILLS

Process Engineering – Process Design, Simulation, Mass and Energy Balances, Equipment Sizing, Pressure Relief Device Sizing, Hydraulic Analysis, Process Optimization, HAZOP, Process Safety Design, Hazardous Area Classification

Simulation & Calculation Tools – Aspen HYSYS, Aspen Plus, Mathcad, Ansys, Autodesk Products

Software Development – Java, Spring Boot, Hibernate, JPA, JUnit, Mockito, Python, REST APIs, Microservices

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Project & Quality Management – PFDs, P&IDs, Equipment Lists, Process Descriptions, Scope of Work, Quality Management System (QMS), Change Control, SOPs

Compliance & Standards – PED, ATEX, GMP, Client Specifications, Regulatory Compliance

PROJECTS

Industrial Process Simulation and Design Study

- Developed detailed process simulations using **Aspen HYSYS** to model chemical reaction kinetics and optimize operating parameters, resulting in a projected 15% increase in yield.
- Generated P&IDs and equipment specification lists for a pilot plant design, adhering to **PED and ATEX** safety standards.
- Collaborated with mechanical engineers to size critical process equipment, including heat exchangers and reactors, based on simulation outputs and hydraulic analysis.

Automated Data Processing Pipeline

- Engineered a **Python**-based data pipeline to automate the ingestion and initial processing of sensor data from industrial control systems, reducing manual data handling time by approximately 40%.
- Developed **REST APIs** using **FastAPI** to expose processed data for consumption by downstream analytics tools, ensuring efficient data retrieval.
- Implemented unit tests for the Python scripts and API endpoints, maintaining code quality and reliability for the data processing workflow.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)